flightPlanner: Flight Routing Engine

An optimal global flight path calculater using different path finding algorithms

Project by

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Project Objectives completed so far

01. Large Scale Graph Construction

Represent flight network of *5,800+ airports and over 1.1 Million flights worldwide.*

02. Advanced Algorithmic Routing

Implemented *Dijkstra, A*, RC-A*, BFS, Yen's Algorithm, Label-Correcting Algorithm,* Two-Phase Router (with & without hub), Contraction Hierarchy.

03. Algorithm Comparison

Speed of finding the **fastest or Cheapest** flights.



Fig: Representation of all flights in China

```
saeedahmedmahin@Saeeds-Mac-mini flightPlanner % rm -rf build
cmake -B build

cmake --build build

./build/flight_router
-- The CXX compiler identification is AppleClang 17.0.0.17000319
-- Detecting CXX compiler ABI info
-- Detecting CXX compiler ABI info - done
-- Check for working CXX compiler: /usr/bin/c++ - skipped
-- Detecting CXX compile features
-- Detecting CXX compile features - done
-- Configuring done (0.3s)
-- Generating done (0.0s)
-- Build files have been written to: /Users/saeedahmedmahin/code/flightPlanner/build
```

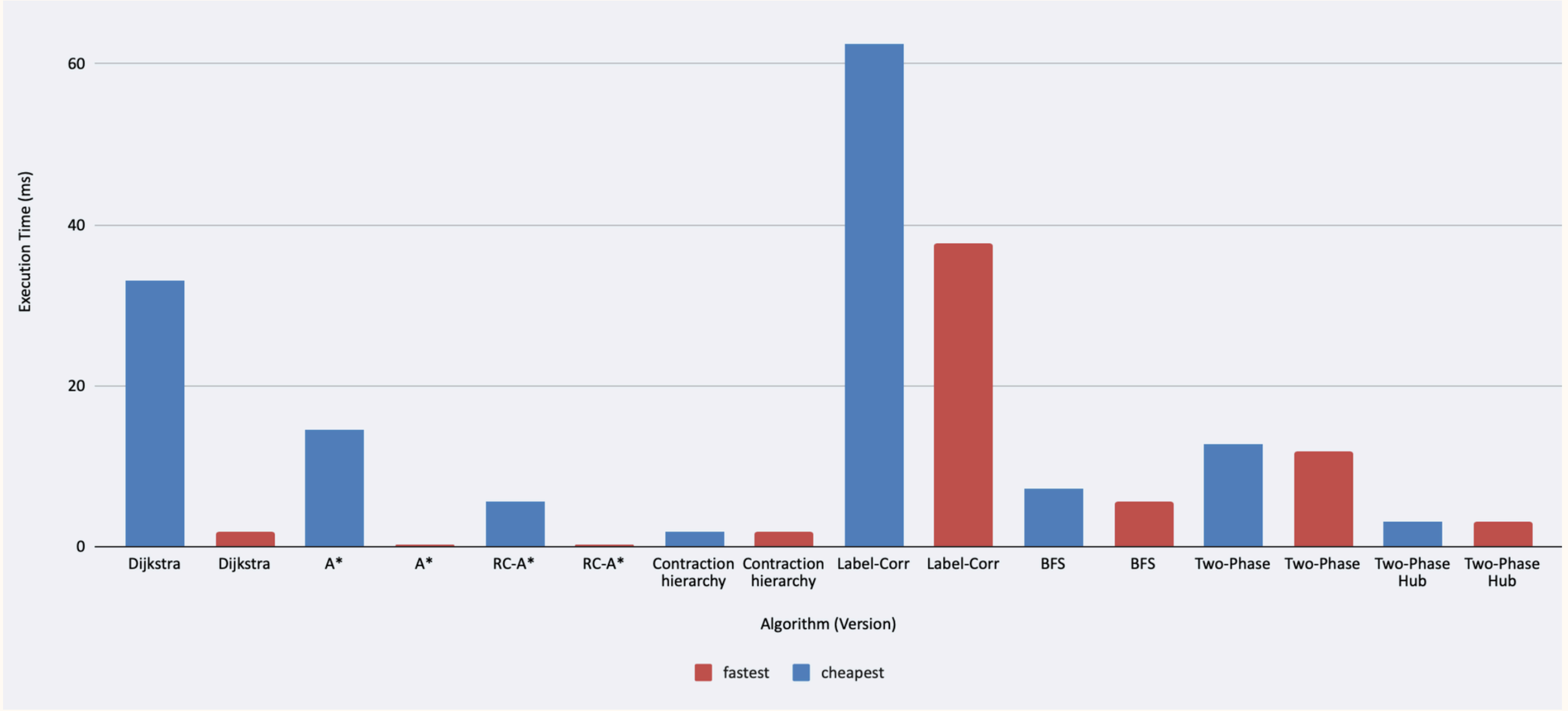
[3]

Calculated Outcome

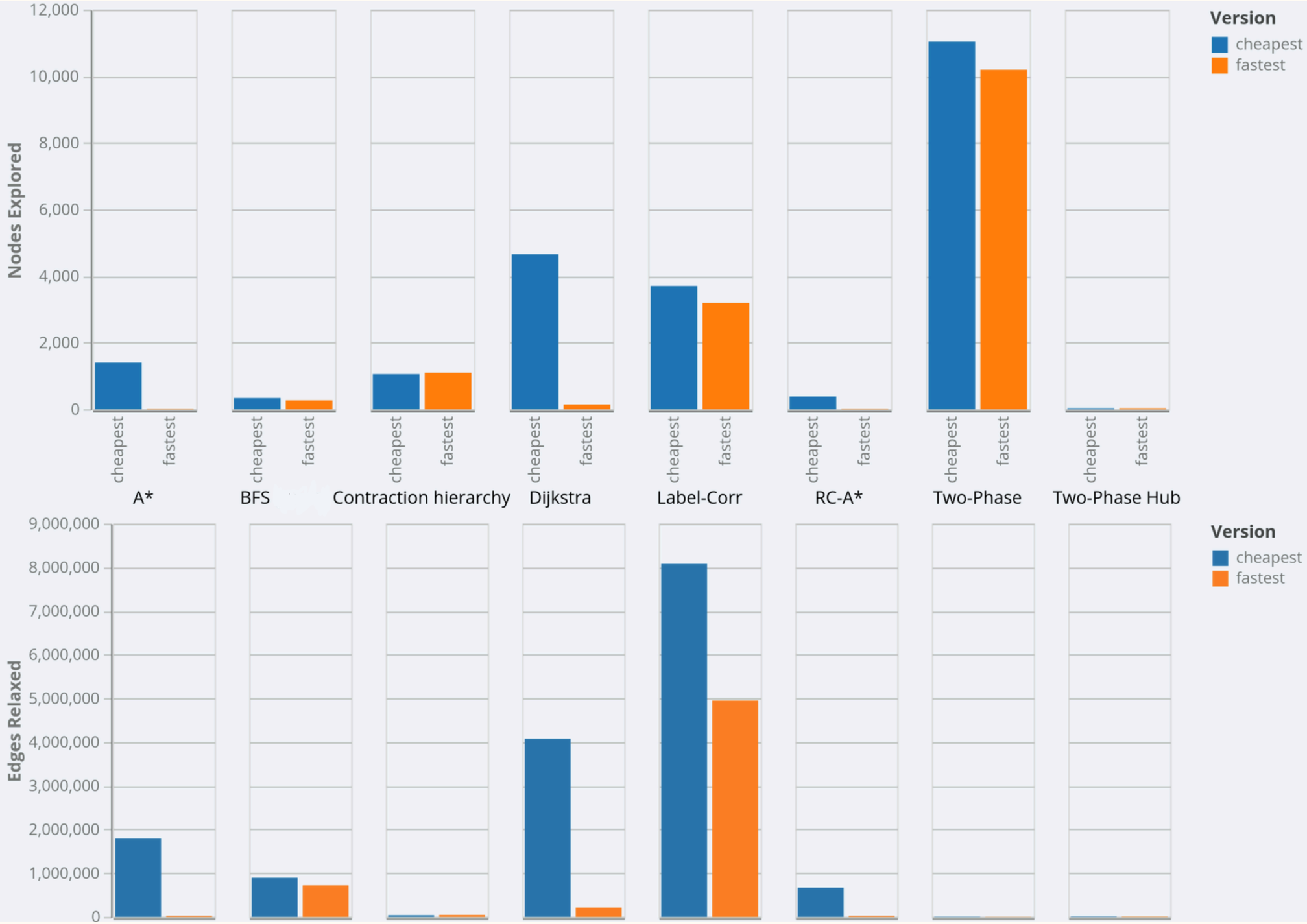
For DAC → JFK

Algorithm	Version	Time	Nodes Explored	Edges Relaxed	Cost	Time Required
Dijkstra	cheapest	32.99 ms	4657	4073669	$644	25h 0m
	fastest	1.78 ms	142	211193	$644	25h 0m
A*	cheapest	14.58 ms	1399	1791962	$644	25h 0m
	fastest	0.27 ms	15	23946	$644	25h 0m
RC-A*	cheapest	5.65 ms	381	667435	$644	25h 0m
	fastest	0.29 ms	15	23946	$644	25h 0m
Contraction hierarchy	cheapest	1.77 ms	1053	38923	$644	25h 0m
	fastest	1.80 ms	1093	45558	$644	25h 0m
Label-Corr	cheapest	62.53 ms	3702	8078340	$644	25h 0m
	fastest	37.62 ms	3189	4950233	$644	25h 0m
BFS (Hops)	cheapest	7.11 ms	334	895260	$729	31h 50m
	fastest	5.50 ms	268	720025	$854	31h 50m
Two-Phase	cheapest	12.74 ms	11039	7715	$644	25h 0m
	fastest	11.90 ms	10196	4163	$644	25h 0m
Two-Phase Hub	cheapest	3.16 ms	33	10833	$644	25h 0m
	fastest	3.01 ms	33	9279	$644	25h 0m
Yen's K-SP	k=5 cheap	505.11 ms	4657	4073669	$644	25h 0m

Visualization of Outcome



[6]


Nodes Explored
12,000
10,000
8,000
6,000
4,000
2,000
0
cheapest
fastest
A*
BFS
Contraction hierarchy
Dijkstra
Label-Corr
RC-A*
Two-Phase
Two-Phase Hub
Version
cheapest
fastest
Edges Relaxed
9,000,000
8,000,000
7,000,000
6,000,000
5,000,000
4,000,000
3,000,000
2,000,000
1,000,000
0
Version
cheapest
fastest

flightPlanner

Project Timeline (*Fortnights*)

